Trigonometric Graphs

Exam code: 0607

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Written by: Naomi C
Reviewed by: Dan Finlay
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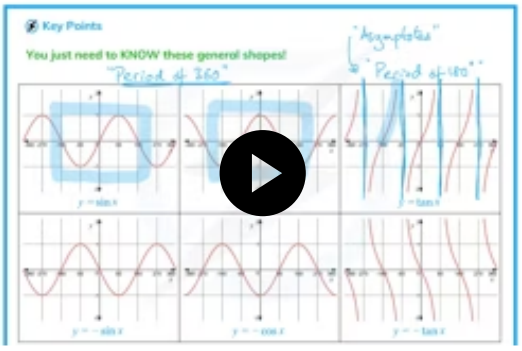


Did this video help you? Yes No

Help me understand

Drawing trig graphs

What are trig graphs?

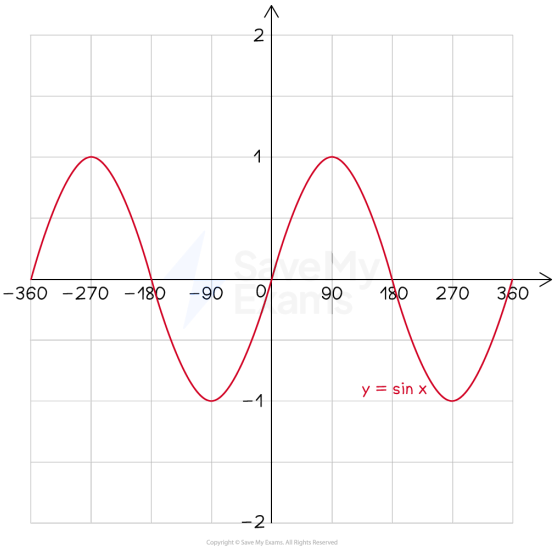
- **Trigonometric (trig) graphs** are the graphs of
  - $y = \sin x$
  - $y = \cos x$
  - $y = \tan x$
- The variable x is like an angle
  - but the angle can now go **beyond acute** to become **obtuse** and **reflex**
    - $0° \leq x \leq 360°$
- Trig graphs have **repeating** (periodic) shapes and **symmetries** that you need to know

How do I draw the graph of y = sin x?

- The graph of $y = \sin x$ is a **wave** that oscillates between **heights of 1 and –1** and **repeats every 360°** (its **period** is 360°)
  - It goes through the **origin**, (0, 0)
  - Plot a point whenever x is a **multiple of 90°**
    - The y coordinates cycle through the numbers 0, 1, 0, -1

x	0°	90°	180°	270°	360°	450°

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

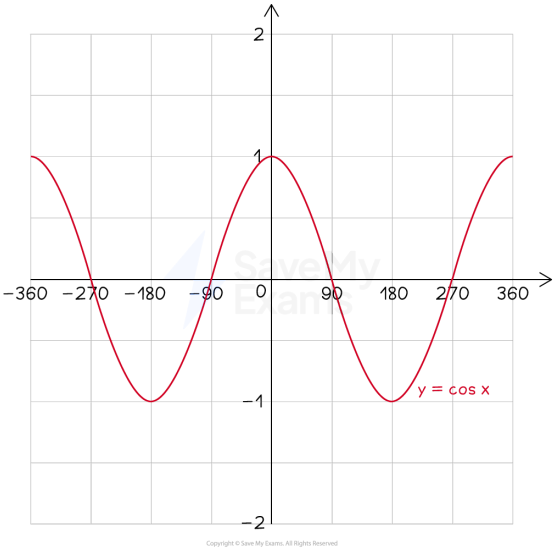


How do I draw the graph of y = cos x?

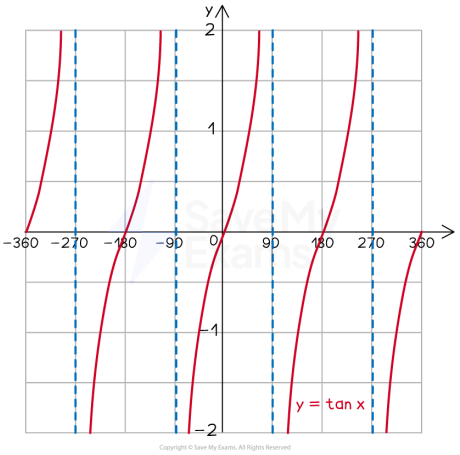
- The graph of $y = \cos x$ is a **wave** that oscillates between **heights of 1 and -1** and **repeats every 360°** (its **period** is 360°)
  - It has a ***y*-intercept of 1**, coordinates (0, 1)
  - Plot a point whenever x is a **multiple of 90°**
    - The y coordinates cycle through the numbers 1, 0, -1, 0

x	0°	90°	180°	270°	360°	450°
y	1	0	-1	0	1	0

- $y = \cos x$ is the same as translating $y = \sin x$ by 90 to the left



- The graph of $y = \tan x$ is **not a wave** but consists of **branches** that **repeat every 180°** (its **period** is 180°)
  - This is **half the period** of $\sin x$ and $\cos x$
- There are **dotted vertical lines** that separate the branches called **asymptotes**
  - These are every 180° at $x = 90°, x = 270°, \ldots$
  - The curve cannot touch these, but get **closer and closer** to them
  - A branch starts down at a height of $-\infty$ and goes up to a height of $+\infty$
- $y = \tan x$ goes through the **origin**, (0, 0)



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